System of Linear Equations

Time limit: 1.00 s
Memory limit: 512 MB

CSES 3154 | Mathematics | https://cses.fi/problemset/task/3154

You are given $n \cdot (m+1)$ coefficients $a_{i,j}$ and b_i which form the following n linear equations:

- $a_{1,1}x_1 + a_{1,2}x_2 + \ldots + a_{1,m}x_m = b_1 \bmod 10^9 + 7$
- $a_{2,1}x_1 + a_{2,2}x_2 + \ldots + a_{2,m}x_m = b_2 \bmod 10^9 + 7$
- $\ldots$
- $a_{n,1}x_1 + a_{n,2}x_2 + \ldots + a_{n,m}x_m = b_n \bmod 10^9 + 7$

Your task is to find any m integers $x_1, x_2, \ldots, x_m$ that satisfy the given equations.

Input

The first line has two integers n and m: the number of equations and variables.

The next n lines each have $m+1$ integers $a_{i,1}, a_{i,2}, \ldots, a_{i,m}, b_i$: the coefficients of the i-th equation.

Output

Print m integers $x_1, x_2, \ldots, x_m$: the values of the variables that satisfy the equations. The values must also satisfy $0 \le x_i < 10^9 + 7$. You can print any valid solution. If no solution exists print only -1.

Constraints

- $1 \le n, m \le 500$
- $0 \le a_{i,j}, b_i < 10^9 + 7$

Example

Input

```
3 3
2 0 1 7
1 2 0 0
1 3 1 2
```

Output

```
2 1000000006 3
```